Result and Discussion: -

Logisim Simulation: -

We started by using Logisim to simulate our Electronic Voting Machine (EVM) design. This helped us see how our system would work without building the actual hardware. In Logisim, we could test different scenarios and check if the buttons, gates, and counters behaved as we planned. It was like a practice run before going to the real hardware.

Verilog Implementation: -

Once we were confident with the Logisim simulation, we moved on to using Verilog. Verilog is like the language that tells the hardware what to do. We translated our design into Verilog code to make the real hardware work just like we tested in Logisim. This step was crucial because it turned our plans into something the actual Electronic Voting Machine could understand and follow.

Output:-

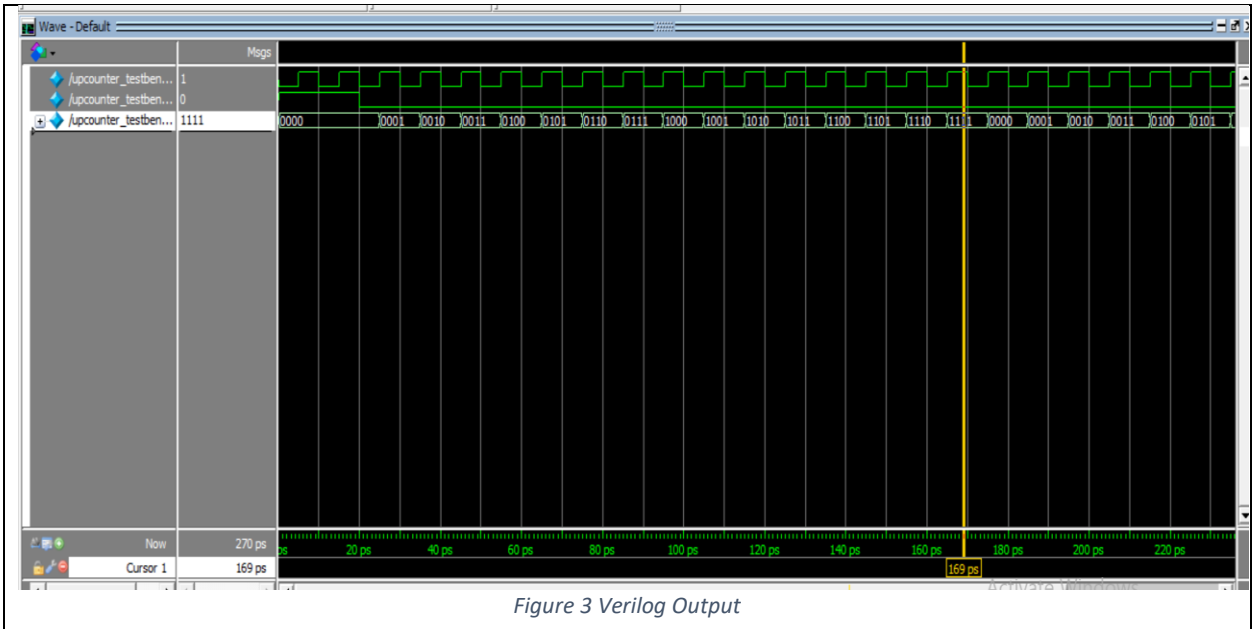


Figure 3 Verilog Output

Output:

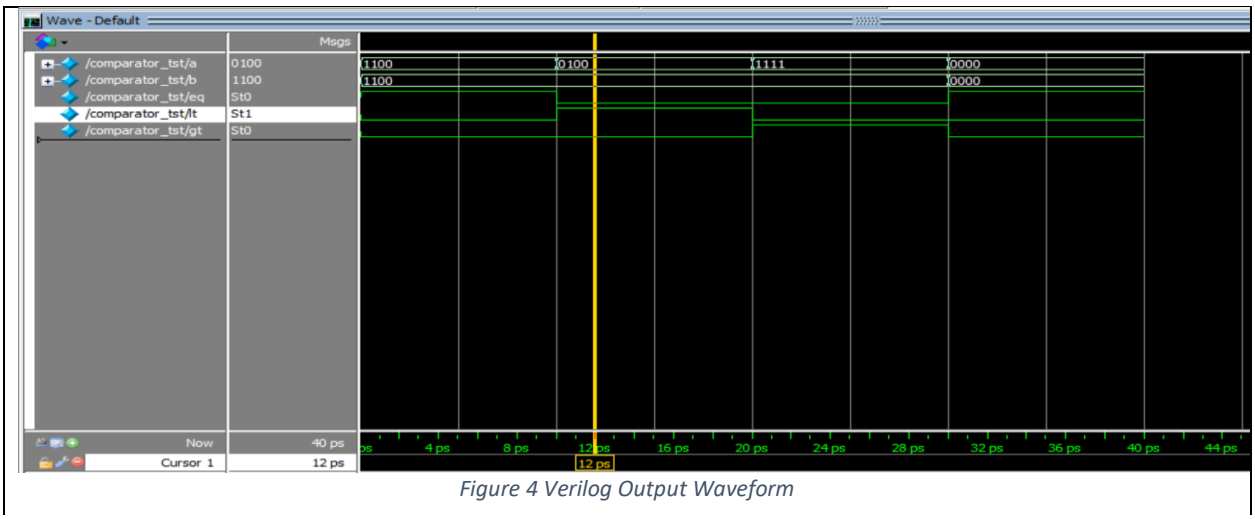


Figure 4 Verilog Output Waveform

Hardware: -

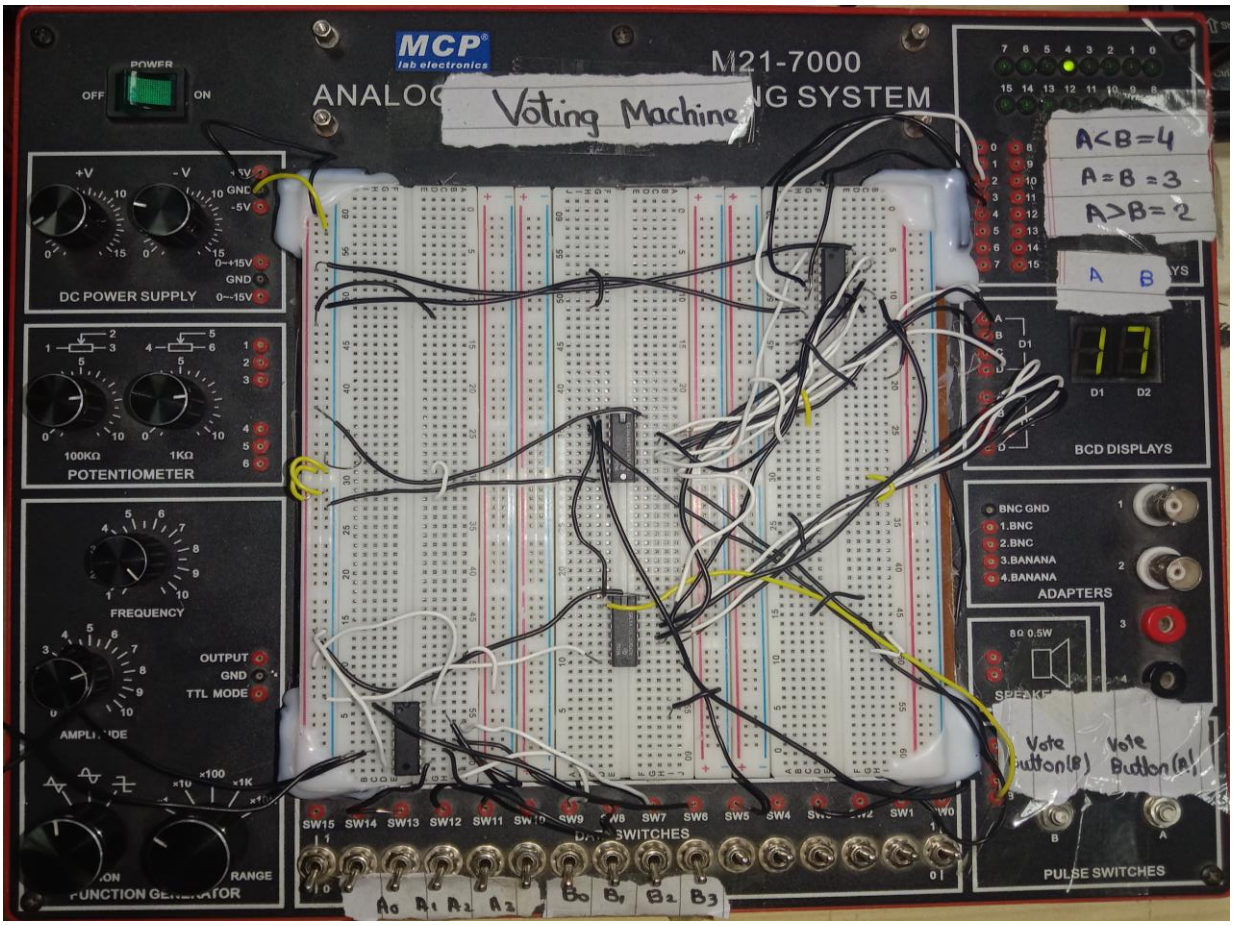


Figure 5 Hardware Implementation

Hardware Results:

Then came the exciting part – building the actual Electronic Voting Machine! We connected all the buttons, gates, and counters, just like in our Logisim simulation and Verilog code. When we tested the real hardware, it matched what we saw in Logisim and what we wrote in Verilog. The buttons worked, the counters counted votes, and the displays showed the results. This showed us that our design, tested in Logisim and translated into Verilog, successfully became a real, working Electronic Voting Machine.

Discussion:

Using Logisim allowed us to play around with our design without worrying about breaking anything. Verilog helped us turn our design into something the hardware could understand. When we put it all together, the hardware matched what we planned and tested. This means our Electronic Voting Machine is reliable and does what it's supposed to do. The combination of Logisim, Verilog, and actual hardware testing was key to making sure our EVM works smoothly and securely.

This journey from simulation to Verilog and finally to real hardware gives us confidence that our Electronic Voting Machine is ready for secure and efficient voting in real-world scenarios.